Vector Factories & Hyperplanes

How Geometry Powers Machine Learning

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A question I hear a lot

How does AI work?

After all, computers are unbelievably stupid!

For example

A human can see exactly what this chunk of Python code means:

```
student = {
    "name": "Alice",
    "score": 97
    "passed": True,
}

print(student)
```

But Python does not say “close enough”:

```
File "example.py", line 4
    "score": 97
             ^
SyntaxError: invalid syntax.
```

Computers do not do what you mean. They do what you say.

So if computers are so dumb...

Then how did we get them to...

- reason about language?
- recognize images?
- generate believable videos from a text description?
- and so on?

The answer you'll hear: we let them learn from data, and miracles ensue.

No miracles. Only math.

Before a machine can learn from words or images, those objects must first be turned into something mathematics can work with.

That is the real foundation for the "miracle."

My goal today: give you some sense of how that works, and what happens next.

- **Vector factories:** how we represent words, pictures, and sounds as vectors in Euclidean space
- **Hyperplanes:** how we use simple geometric boundaries in that space to sort, separate, and make decisions

A first geometric question

Suppose we observe the weather every day and record two features:

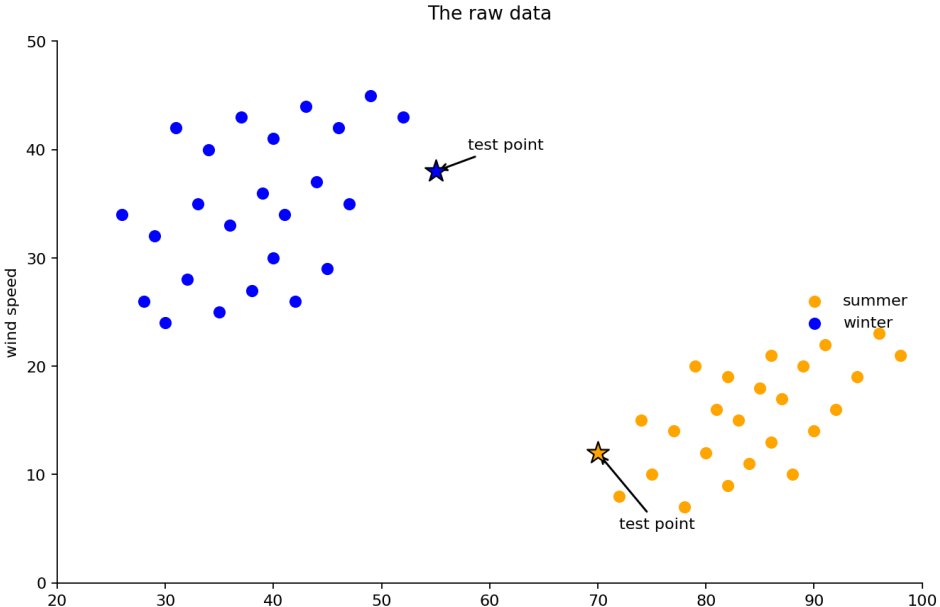
- x_1 = temperature
- x_2 = max wind speed

Can we draw a boundary in this 2D feature space that separates:

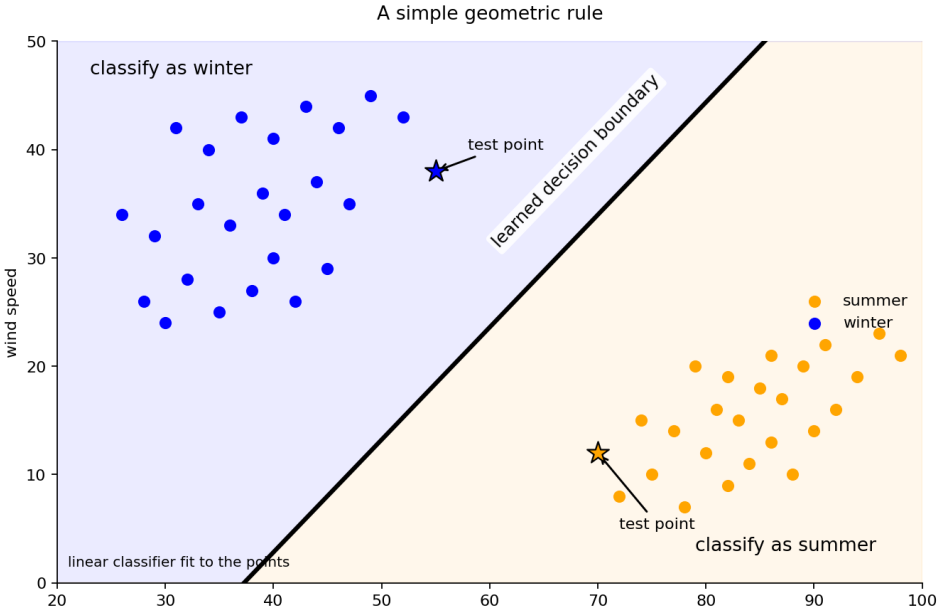
- summer days ($y = 1$), versus
- winter days ($y = 0$)?

This is a simple **classification** problem.

The raw data


The raw data
wind speed
test point
test point
summer
winter
0
10
20
30
40
50
20
30
40
50
60
70
80
90
100

The linear decision boundary



A simple decision boundary

A line like this is called a **linear decision boundary**.

In higher dimensions, the same idea becomes a plane, or more generally a **hyperplane**.

For a point $x \in \mathbb{R}^d$, we compute

$$w^\top x + b = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d + b.$$

The sign of this number tells us which side of the boundary x lies on.

Why do we like linear rules?

- **Simple:** easy to describe and interpret
- **Trainable:** easy to differentiate and so easy to optimize
- **Fast:** modern hardware is extremely good at linear algebra, even in super high dimensions

A problem a line cannot solve

Imagine a drone trying to land on a circular charging pad.

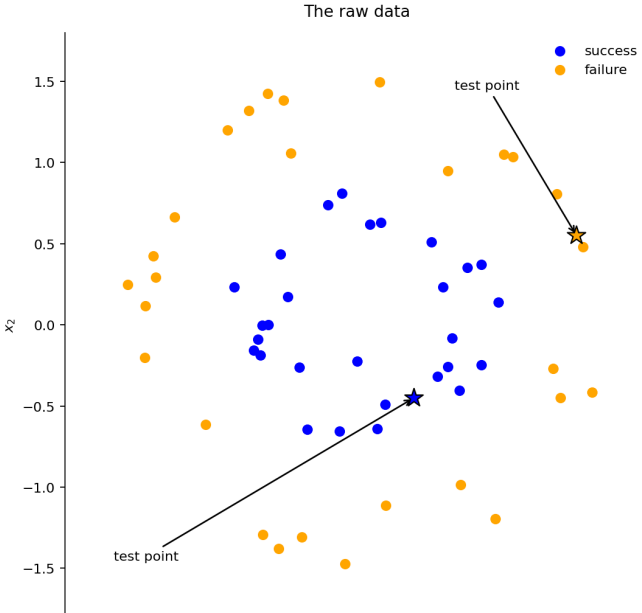
For each landing attempt, record

- x_1: horizontal error from the center
- x_2: vertical error from the center

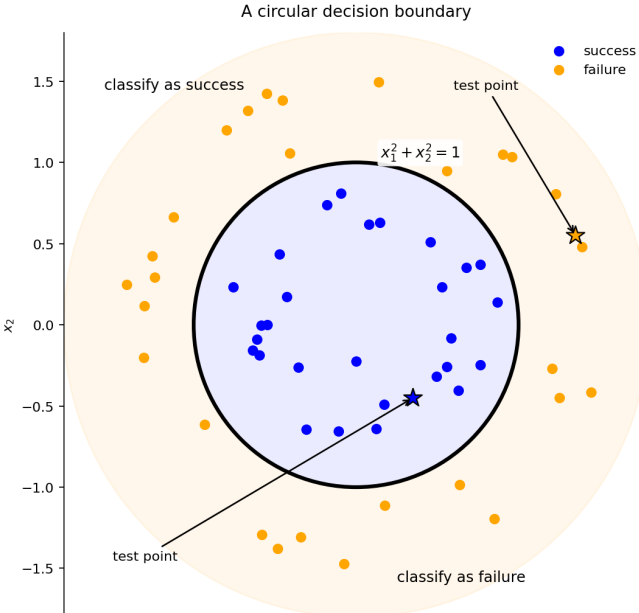
We want a decision rule:

- **success** if the landing is close enough to the center
- **failure** if it lands too far away

The raw data


The raw data
success
failure
test point
test point
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
x_2

The right boundary in this space



Problem: not linear

The natural rule is based on radial distance from the center:

$$x_1^2 + x_2^2 > 1$$

This is a nonlinear decision boundary. No line separates the classes.

We like this a lot less:

- Much harder to search for good nonlinear boundaries in high-dimensional feature spaces
- Much harder to optimize and deploy efficiently at scale

Solution: add a new feature

Introduce a third coordinate:

$$x_3 = x_1^2 + x_2^2.$$

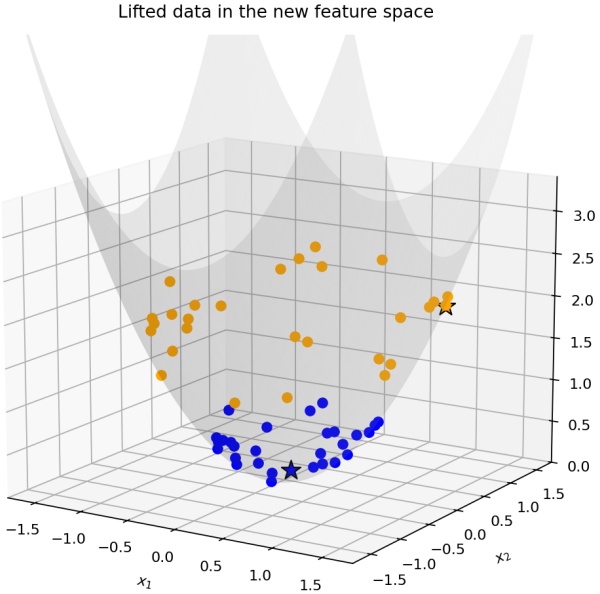
Now lift each point from

$$\mathcal{X} = (x_1, x_2) \quad \text{to} \quad \mathcal{X}' = (x_1, x_2, x_3).$$

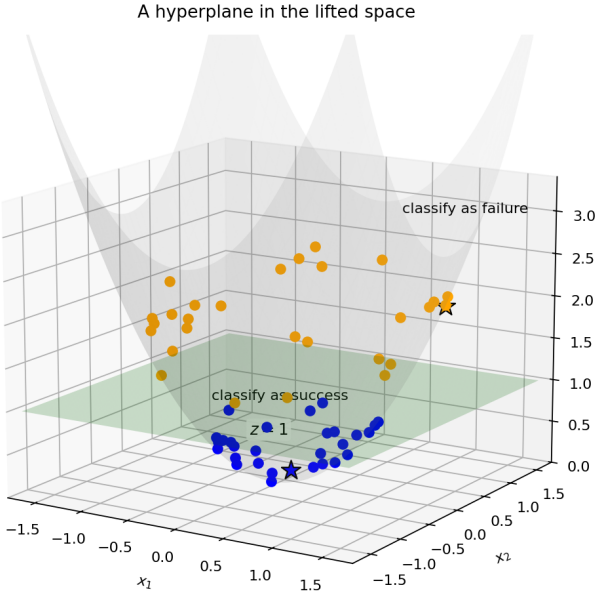
In the lifted space:

- Points near the center have small x_3.
- Points farther out have larger x_3.

The points in lifted space



A separating hyperplane



The lesson of the lifting trick

The hyperplane was not the hard part.

The hard part was inventing the right new feature:

$$x_3 = x_1^2 + x_2^2.$$

Once we had that feature, the separation became easy.

But in real problems, the right features are usually not obvious.

The real bottleneck

For words, images, and sounds, we usually do **not** know in advance

- which coordinates matter
- which transformations make the data separable
- which geometric directions will be useful

That is the hard part: building the right representation.

So where do good coordinates come from?

In machine learning, we need systematic ways to construct good coordinates:

- for making similar things land near each other
- for making useful patterns visible
- for making simple decision rules, like hyperplanes, actually work

It's not obvious where to start if our data points aren't numbers!

That's the harder half of today's story: **vector factories,** or ways of turning messy objects into useful coordinates.

Many data types, many vector factories

There is no single universal recipe for turning data into vectors.

Different kinds of objects usually need different pipelines:

- text
- images
- audio
- video

All of them aim at the same goal: turn messy objects into Euclidean vectors, so that geometry becomes useful.

Today, we'll focus on **text**.

Let's play a game

I'm thinking of a person, place, or thing.

You may ask only yes-or-no questions.

Can you guess what I'm thinking of?

My word

The Muppet version

https://www.youtube.com/embed/PAOKOAtti1A?start=17&end=105

What have we done here?

Let $\mathcal{X}$ be our original space of objects (people/places/things). Our "vector factory" consists of N binary features

$$f_1, \dots, f_N, \qquad f_i : \mathcal{X} \to \{0, 1\}.$$

Each feature asks one yes/no question about the input word x. Together they define the mapping

$$\Phi : \mathcal{X} \to \{0, 1\}^N, \qquad \Phi(x) = (f_1(x), \dots, f_N(x)).$$

We have *embedded* our original space $\mathcal{X}$ into the binary cube $\{0, 1\}^N$.

In the "Muppet Christmas Carol" coordinates

question	Scrooge
vegetable	0
mineral	0
animal	1
found on a farm	0
found in the city	1
pulls a taxi cab	0
bear	0
unwanted creature	1
a rat, leech, or cockroach	0

So we can represent "Scrooge" by the vector $(0, 0, 1, 0, 1, 0, 0, 1, 0)$

Why the binary cube is awkward

In “20 questions”, our embedding space is the binary cube. Two problems appear immediately.

First, some questions don’t have binary answers.

- Is Scrooge a bear? Not literally, but sort of.
- Is he an unwanted creature? Seemingly so... but not to the woman he almost married as a young man.

Dickens knew this

> *"I have found it out! I know what it is, Fred! I know what it is!" "What is it?" cried Fred.*
> *"It's your Uncle Scro-o-o-o-oge!"*
> *Which it certainly was. Admiration was the universal sentiment, though some objected that the reply to "Is it a bear?" ought to have been "Yes;" inasmuch as an answer in the negative was sufficient to have diverted their thoughts from Mr. Scrooge, supposing they had ever had any tendency that way.*

— Charles Dickens, A Christmas Carol

Why the binary cube is awkward

Second, we'd like to use math to:

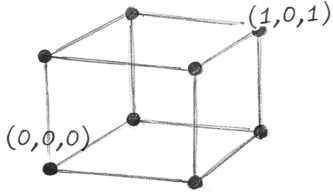
- compare word meanings
- average similar examples
- let data discover patterns and directions

But $\{0,1\}^N$ is nothing but corners.

- averages leave the space
- there is no natural notion of "between" (not convex)

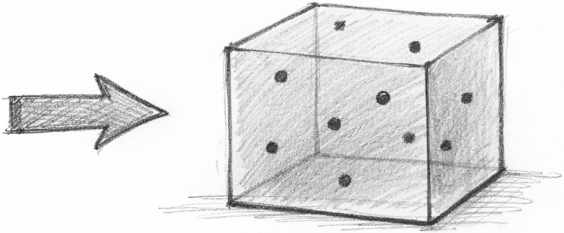
So basic math operations push us outside $\{0,1\}^N$. We'd prefer a space that supports both *meaning* and *math*.

Better



binary cube

only corners



convex hull

Scrooge in real-valued Muppet coordinates

question	Scrooge
vegetable	0.10
mineral	0.00
animal	0.75
found on a farm	0.00
found in the city	0.95
pulls a taxi cab	0.00
bear	0.60
unwanted creature	0.85
a rat, leech, or cockroach	0.15

More words in real-valued Muppet coordinates

question	Scrooge	Rafael Nadal	Winnie-the-Pooh	New York City
vegetable	0.10	0.00	0.15	0.00
mineral	0.00	0.10	0.00	0.30
animal	0.75	0.85	0.90	0.15
found on a farm	0.00	0.05	0.20	0.00
found in the city	0.95	0.60	0.10	1.00
pulls a taxi cab	0.00	0.00	0.00	0.30
bear	0.60	0.30	0.90	0.00
unwanted creature	0.85	0.05	0.00	0.20
a rat, leech, or cockroach	0.15	0.00	0.00	0.40

Another problem

20 questions is adaptive.

Later questions depend on earlier answers.

- If the answer to “animal” is yes, I ask one kind of next question.
- If the answer to “city” is yes I ask a very different one.
- Different targets lead to different paths through the game.

Another problem

But an embedding needs a **fixed coordinate system**.

Every word must be mapped into the same space:

$$\Phi(w) \in \mathbb{R}^d$$

with the same d coordinates for every word.

Muppet coordinates aren't great for this

coordinate	concerto	algorithm	rainbow
vegetable	0	0	0
mineral	0	0	0
animal	0	0	0
found on a farm	0	0	0
found in the city	0	0	0
pulls a taxi cab	0	0	0
bear	0	0	0
unwanted creature	0	0	0
a rat, leech, or cockroach	0	0	0

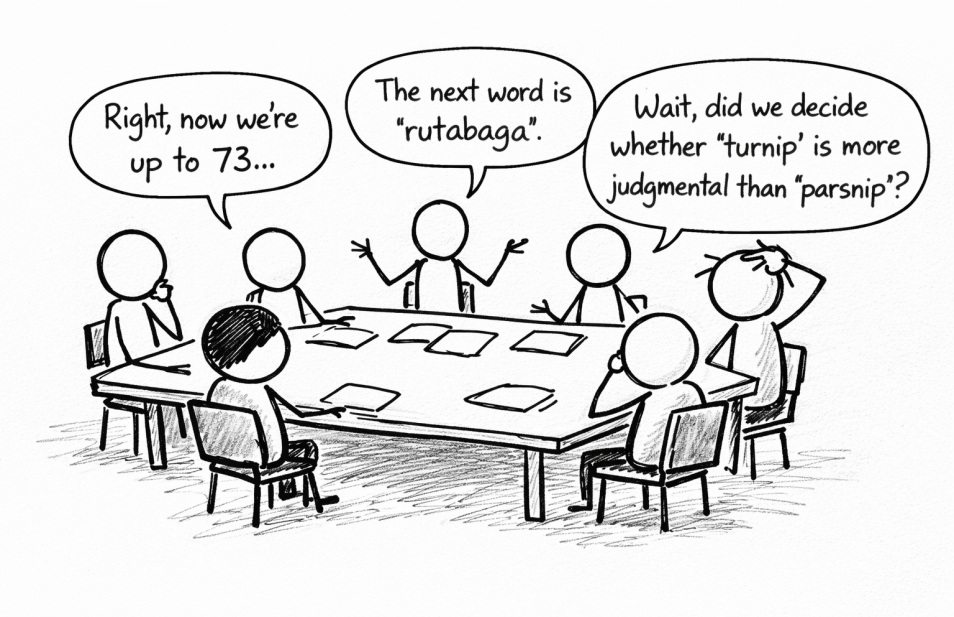
Whole regions of meaning are invisible to these chosen questions.

The real challenge

We need a fixed set of, say, 500 questions that is sufficiently rich to encode the meaning of *all* words.

How?

Option 1: BOGSAT



Option 2: Let the computer choose

This is precisely what's done in practice.

But what kinds of questions could a computer even choose?

Can you guess the missing word?

- He wiped a streak of ____ off the corner of his mouth.
- Could you pass the ____ for the fries?
- The burger came with lettuce, onion, pickles, and ____.
- He got ____ all over the plate when he squeezed the bottle too hard.
- There was a half-empty bottle of ____ next to the salt and pepper.

Notice what you just did.

You did not identify the word as "ketchup" by answering philosophical questions like "is it judgmental?"

You identified it from the company it keeps.

What questions can a computer ask?

A computer does not understand "ketchup" the way we do.

But if you give it access to every English sentence ever digitized, it can ask statistical questions like:

- Does this word often appear near **fries**?
- Does it often appear near **burger**?
- Does it often appear near **bottle**?
- Does it often appear near **salt** or **pepper**?
- Does it often appear in the context **pass the ____**?

These are not philosophical questions. They are questions about **co-location statistics**.

From contexts to coordinates

For each word, we record how often it appears near context words like

$$\text{fries},\ \text{burger},\ \text{bottle},\ \text{salt},\ \text{pepper}, \ldots$$

So instead of hand-designed features, we get data-driven coordinates:

$$\text{ketchup} \mapsto (\text{freq near fries},\ \text{freq near burger},\ \text{freq near bottle}, \ldots)$$

Now the coordinates come from usage patterns in text.

But which context questions?

We still face a problem:

- near **fries**?
- near **burger**?
- near **bottle**?
- near **diner**?
- near **mustard**?
- near **tomato**?
- near **plate**?

If we hand-design context questions, we're back to BOGSAT.

The modern idea

Don't write the questions down explicitly.

Instead:

- assign each word an unknown vector.
- look at nearby words in real sentences: the *context window*
- train a model to predict a missing or next word: "Could you pass the ____ for the fries?"

If the model gets good at using vectors of nearby words to predict the missing word, those vectors must be capturing useful information.

Key ingredient 1: word vectors

Let $\mathcal{V}$ be the vocabulary.

For each word $w \in \mathcal{V}$, we introduce an unknown vector

$$v_w \in \mathbb{R}^d.$$

These vectors are not given to us in advance.

They are parameters to be learned from data.

Key ingredient 2: a prediction function

We also must train a prediction function

$$f_\theta : \mathbb{R}^{md} \to \mathbb{R}^{|\mathcal{V}|},$$

where

- $m =$ number of words in the context window
- $d =$ dimension of each word vector
- $|\mathcal{V}| =$ size of the vocabulary

The input is the sequence of vectors $v_1, \ldots, v_m$ for the m nearby words.

The output is a score for every possible target word in English.

More about this function...

Suppose

- each word vector has dimension $d = 100$
- we look at $m = 10$ nearby words
- the vocabulary has size $|\mathcal{V}| = 50{,}000$

Then the model is a function

$$f_\theta : \mathbb{R}^{10\times 100} \to \mathbb{R}^{50{,}000}.$$

Input: the 10 nearby words, each represented by a vector in $\mathbb{R}^{100}$

Output: 50,000 scores, one for each possible word in the vocab.

The machine learning problem

We then train **both**

- the word vectors v_w
- the prediction function's parameters θ

The model rewards combinations of θ and the v_w's that generate:

- high scores for the right word: "Could you pass the **ketchup** for the fries?"
- low scores for wrong words: "Could you pass the **Immanuel Kant** for the fries?"

Publicly disclosed frontier scale

Toy example:

$$10 \text{ words} \times 100 \text{ dimensions} \quad \longrightarrow \quad 50{,}000 \text{ possible outputs}$$

Publicly disclosed real systems (Llama 4, Gemma 3, OpenAI gpt-oss)

- 128K token context windows are now standard in major open models
- output vocabularies are often around 200K to 260K tokens
- some frontier systems have context windows up to 10 million tokens

Same basic game, wildly larger scale, e.g.

$$f_\theta : \mathbb{R}^{128{,}000 \times 4096} \longrightarrow \mathbb{R}^{260{,}000}$$

What might f_θ even look like?

Usually not one giant formula.

Instead, think of a composition

$$f_\theta = f^{(L)} \circ f^{(L-1)} \circ \cdots \circ f^{(1)}$$

of many simpler functions.

A typical layer looks like

$$x \mapsto \sigma(Ax + b),$$

where A is a matrix, b is a bias vector, and σ is a simple nonlinearity.

Linear rules again

So even a huge model is built from familiar ingredients:

- matrix-vector multiplication
- affine maps
- simple nonlinearities
- repeated many times

Linear decision rules have not disappeared.

They have been layered, recombined, and lifted into richer spaces.

The moral

The real trick is usually not the prediction rule (e.g. a linear boundary).

It is finding the representation in which a simple rule becomes possible.

That is the partnership at the heart of modern machine learning:

- **Vector factories** build the space
- **Hyperplanes** make the space useful

Thank you all!

Bonus: the basic training objective

Suppose a training sentence is

$$w_1, w_2, \dots, w_T.$$

An autoregressive model factors its probability as

$$P(w_1, \dots, w_T) = P(w_1) \prod_{t=1}^{T-1} P(w_{t+1} \mid w_1, \dots, w_t).$$

So one length-T sequence gives us $T - 1$ prediction tasks. We train the model on **trillions** of such sequences.

Bonus: the loss function

Training tries to make the observed next word likely.

So we maximize the log-likelihood

$$\sum_{t=1}^{T-1} \log P_\theta(w_{t+1} \mid w_1, \dots, w_t),$$

or equivalently minimize the negative log-likelihood

$$\mathcal{L}(\theta) = -\sum_{t=1}^{T-1} \log P_\theta(w_{t+1} \mid w_1, \dots, w_t).$$

Bonus: from words to vectors

Each vocabulary item w gets a trainable embedding vector

$$v_w \in \mathbb{R}^d.$$

So an input sequence

$$w_1, \dots, w_T$$

becomes a sequence of vectors

$$v_{w_1}, \dots, v_{w_T}.$$

These vectors are learned jointly with the rest of the model.

Bonus: contextualization

Those raw word vectors are not yet context-sensitive.

So we pass them through many "self-attention" blocks:

$$(v_1, \dots, v_T) \longmapsto (h_1^{(1)}, \dots, h_T^{(1)}) \longmapsto \cdots \longmapsto (h_1^{(L)}, \dots, h_T^{(L)}).$$

Each layer updates every vector using information from the surrounding words.

Bonus: what self-attention is doing

At position t, the model forms a new vector by mixing information from earlier positions:

$$h'_t = \sum_{s \leq t} \alpha_{ts} \, M h_s.$$

Here

- the coefficients α_{ts} say how much position t attends to position s
- the matrix M linearly transforms the vectors before mixing

So each new vector is a weighted combination of context-aware linear transforms of earlier vectors.

Bonus: from the final layer to next-word scores

After the last self-attention block, we have final vectors

$$h_1, \dots, h_T.$$

To predict the next word after the context $w_1, \dots, w_T$, we use the last vector h_T and compute

$$z = W_{\text{out}} h_T + b,$$

where

$$z \in \mathbb{R}^{|\mathcal{V}|}.$$

So we get one score for every possible completion word.

Bonus: from scores to probabilities

The score vector

$$z \in \mathbb{R}^{|\mathcal{V}|}$$

is turned into a probability distribution by softmax:

$$P_\theta(w \mid w_1, \dots, w_T) = \frac{e^{z_w}}{\sum_{u \in \mathcal{V}} e^{z_u}}.$$

So the model assigns a probability to every possible next word.

Bonus: what is learned?

Training adjusts both

- the embedding vectors v_w
- the internal parameters of the self-attention blocks
- the final output map W_{out}, b

so that real next words get high probability and implausible completions get low probability.

Bonus: the whole pipeline

$$
\begin{aligned}
w_1, \dots, w_T &\longrightarrow v_{w_1}, \dots, v_{w_T} \\
&\longrightarrow h_1, \dots, h_T \\
&\longrightarrow W_{\text{out}} h_T + b \\
&\longrightarrow P_\theta(\text{next word} \mid w_1, \dots, w_T)
\end{aligned}
$$

So the model learns both

- a vector space for words
- and a giant compositional function acting on sequences of those vectors

Bonus: same story, more machinery

Even here, the basic picture has not changed.

We still have:

- vectors representing words
- linear algebra doing most of the work
- simple nonlinearities and normalization
- a final scoring rule over possible outputs

The machinery is elaborate.

The geometric story is the same.